Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
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Practice Exam — Week 6: Fourier Theory

Duration: 60 minutes · No calculator · Answer on paper

Name: ______________________ Surname: ______________________

SCIPER (Student Card Number): ______________________

Exercise	**Points**
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2.

Problem 1 (5 points)

Throughout this problem $g \in L^1$ has continuous-time Fourier transform

$$G(f) = \int_{-\infty}^{\infty} g(t)\, e^{-2\pi i f t}\, dt,$$

and for a derived function h we write H for its transform. We use the four elementary Fourier-transform properties (conjugation, scaling, shift, linearity).

(a) Prove the *conjugation* and *time-shift* rules: if $h(t) = \overline{g(t)}$ then $H(f) = \overline{G(-f)}$, and if $h(t) = g(t+\tau)$ then $H(f) = G(f)\, e^{2\pi i f \tau}$. (2 pts)

(b) Combining scaling and shifting, show that the *modulated* signal $h(t) = g(\alpha t)\, e^{2\pi i f_0 t}$ (with $\alpha \neq 0$, $f_0 \in \mathbb{R}$ fixed) has transform

$$H(f) = \frac{1}{|\alpha|}\, G\!\left(\frac{f - f_0}{\alpha}\right).$$

State in one sentence what each of the two operations (scaling by α, modulation by $e^{2\pi i f_0 t}$) does to the spectrum. (2 pts)

(c) Use part (a) to show that if g is *real and even* (so $g(t) = \overline{g(t)} = g(-t)$), then its transform G is real and even. (1 pt)

Problem 2 (5 points)

Recall the continuous-time convolution of $g, h \in L^1$,

$$(g * h)(t) = \int_{-\infty}^{\infty} g(t-y)\, h(y)\, dy,$$

and that upper-case letters denote Fourier transforms.

(a) Prove the *convolution theorem*: if $u = g * h$ then $U = G \cdot H$. State explicitly the Fubini step and why the hypotheses $g, h \in L^1$ justify it. (3 pts)

(b) *(Filtering.)* A signal $x \in L^1$ is passed through a linear filter with impulse response $a \in L^1$, producing $y = a * x$. Using part (a), give Y in terms of X and the transfer function A. If the filter is the running-average kernel $a(t) = \frac{1}{2T}\, \mathbf{1}_{\{|t| \le T\}}$, compute $A(f)$ in closed form and verify $A(0) = 1$. (2 pts)

Problem 3 (5 points)

This problem concerns the finite-data Fourier transform, the Dirichlet kernel, and aliasing. Take sampling interval $\Delta = 1$ and observation times $t = 0, 1, \ldots, n-1$.

(a) The transform of the finite rectangular window is the **Dirichlet kernel**

$$D_n(f) = \sum_{t=0}^{n-1} e^{-2\pi i f t}.$$

Sum this finite geometric series and show that

$$D_n(f) = \frac{\sin(n\pi f)}{\sin(\pi f)}\, e^{-\pi i (n-1) f}, \qquad \text{so} \qquad |D_n(f)| = \left|\frac{\sin(n\pi f)}{\sin(\pi f)}\right|.$$

(2 pts)

(b) Show that $|D_n(f)|$ vanishes exactly at the nonzero **Fourier frequencies** $f_k = k/n$, $k = 1, \ldots, n-1$, while $D_n(0) = n$. (This is why the finite Fourier transform of a pure sinusoid sitting exactly on a Fourier frequency contributes nothing at the other Fourier frequencies.) (1.5 pts)

(c) *(Aliasing / Nyquist.)* Let $g \in L^1$ have continuous-time Fourier transform $\mathcal{G}(f)$, and let G denote the transform of the sampled sequence $\{g(t)\}_{t \in \Delta\mathbb{Z}}$. Recall the aliasing identity

$$G(f) = \sum_{k \in \mathbb{Z}} \mathcal{G}\left(f + \frac{k}{\Delta}\right), \qquad |f| \le \frac{1}{2\Delta}.$$

Suppose $\mathcal{G}$ is band-limited triangular, $\mathcal{G}(f) = (1 - |f|)\, \mathbf{1}_{\{|f| \le 1\}}$. For $\Delta = 1$ (Nyquist frequency $\frac{1}{2}$) compute the aliased transform $G(f)$ on $|f| \le \frac{1}{2}$, and state whether aliasing has occurred and why. (1.5 pts)

Problem 4 (5 points)

(Revision of Week 5: differencing & SARIMA.) Consider the signal-plus-noise model $X_t = \mu_t + Y_t$, where μ_t is a deterministic component and $\{Y_t\}$ is a zero-mean stationary process with autocovariance sequence γ_τ. Recall $\nabla = I - B$, $\nabla_{(s)} = I - B^s$.

(a) Let $\mu_t = a + bt + s_t$ with a *monthly* season $s_{t+12} = s_t$. Compute $\nabla \nabla_{(12)} X_t$ explicitly as a linear combination of the Y's, verify its mean is 0, and explain in one sentence why it is weakly stationary. (2 pts)

(b) Now let $\{X_t\}$ be the multiplicative seasonal process $\mathrm{SARMA}(0,1) \times (1,0)_{12}$,

$$\big(1 - \Phi\, B^{12}\big) X_t = \big(1 - \theta B\big)\varepsilon_t, \qquad |\Phi| < 1.$$

By inverting the seasonal AR factor, obtain the MA(∞) weights ψ_j in $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$ (closed form for ψ_{12l} and ψ_{12l+1}; all other $\psi_j = 0$). (2 pts)

(c) Using $\gamma_\tau = \sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$, compute γ_0 and γ_1 in closed form, and evaluate $\rho_1 = \gamma_1/\gamma_0$ for $\theta = \frac{1}{2}$. (1 pt)